

LECTURE 3

Supervised Learning Part 2: Regression Algorithms & Sensor Fusion

OLS • Ridge • Lasso • SVR • XGBoost Regression • Kalman Filter • ML-enhanced Sensor Fusion

Case A: mission-duration forecast (N = 6,200, 18 features) • Case B: 3-sensor radar + IR + ELINT fusion (5 targets)

AK DHAMIJA, Researcher & Instructor

AI & Machine Learning

dhamija.ak@gmail.com • 9811453163

Regression + fusion: two case studies, one unified theme

Learning objectives

1. Derive OLS in closed form
2. Ridge (L2) & Lasso (L1) penalties
3. Why regularisation prevents overfitting
4. SVR with ε-insensitive loss
5. Apply XGBoost to regression
6. Select metrics (MSE/RMSE/MAE/R^2)
7. Derive Bayesian prediction intervals
8. Kalman predict–update cycle
9. Early, late and hybrid fusion
10. Quantify ML-vs-Kalman benefit

Data pitfall: regression is especially vulnerable to outliers — one extreme battle record can pull OLS coefficients. Plot residuals; report RMSE with R^2.

Case A: mission duration

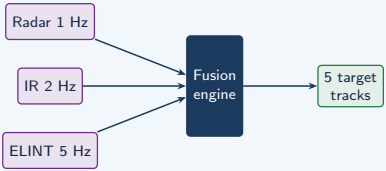
Predict time-to-mission-completion (h) from pre-mission parameters. $N = 6{,}200$ exercise records; split 70/15/15; 18 features: terrain (5 cat.), weather (4), force size, armour count, air assets, OPFOR density, EW intensity, supply level, mission type (6), ROE (4), distance, elevation change, comms quality, morale, days since resupply, temperature, fog-of-war, enemy experience.



regression: $\hat{y} \in \mathbb{R}$

Case B: 3-sensor fusion

Track 5 simultaneous airborne targets fusing 3 independent sensors. **S1** ground radar (range, bearing, speed); **S2** IR (bearing, elevation, temperature); **S3** ELINT (frequency, PRF, bearing). Different update rates (1, 2, 5 Hz), noise ($\sigma^2_{\text{radar}} \gg \sigma^2_{\text{ELINT}}$), overlap (bearing common to all), missing data (S2 blocked by cloud, 20% gaps).



MODEL, LOSS AND DERIVATION

$$y = Xw + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I)$$

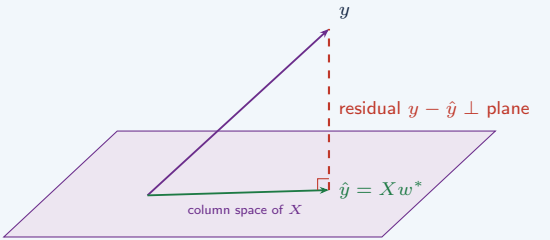
$$L(w) = \|y - Xw\|^2 = \sum_{i=1}^{N} (y_i - w^\top x_i)^2$$

$$\frac{\partial L}{\partial w} = -2X^\top (y - Xw) = 0 \;\Rightarrow\; X^\top y = X^\top X w$$

$$\boxed{w^* = (X^\top X)^{-1} X^\top y}$$

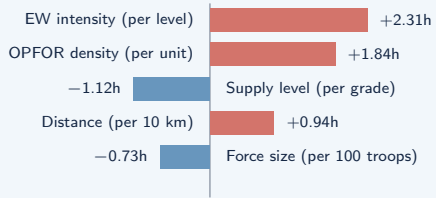
- $X \in \mathbb{R}^{N \times d}$ ($N = 4{,}340$, $d = 18$), $w \in \mathbb{R}^d$, $y \in \mathbb{R}^N$
- **Gauss–Markov:** linearity $E[y|X] = Xw$; no multicollinearity; homoscedastic $\mathrm{Var}(\varepsilon_i) = \sigma^2$; independent errors $\Rightarrow$ OLS is **BLUE**

Geometry: projection onto the column space of X



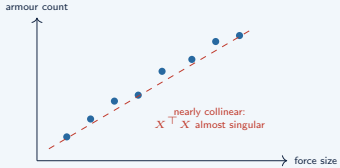
Case A (OLS, $N = 4{,}340$): train $R^2 = 0.738$ (RMSE 3.97 h); val 0.712 (4.18 h); **test 0.710 (4.22 h)**. Target range [2.1, 48.6] h.

Top-5 coefficient magnitudes (effect on mission duration, hours)



Interpretability: EW intensity is the strongest predictor of mission duration — aligned with doctrine.

Collinearity check: force size and armour count have **VIF = 8.4** (borderline), so coefficients may be distorted. Solution: ridge regression (next slide).



VARIANCE INFLATION FACTOR

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

RIDGE OBJECTIVE AND CLOSED FORM

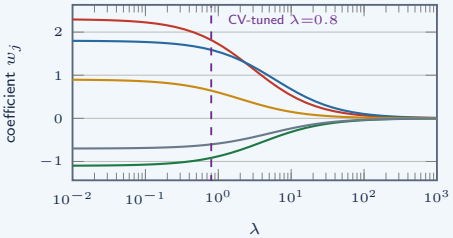
$$L_{\text{Ridge}}(w) = \|y - Xw\|^2 + \lambda\|w\|^2$$

$$w^*_{\text{Ridge}} = (X^\top X + \lambda I)^{-1} X^\top y$$

- OLS overfits when features are collinear or d is large relative to N
- $X^\top X + \lambda I$ is **always invertible** for $\lambda > 0$, even if X is rank-deficient
- $\lambda = 0$: OLS; $\lambda \to \infty$: $w \to 0$; tuned λ: bias ↑ but variance ↓

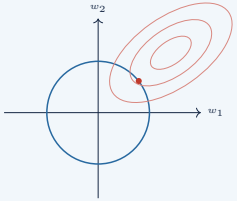
Case A ($\lambda = 0.8$, 5-fold CV): test $R^2 = 0.738$, RMSE 3.91 h (vs OLS 0.710 / 4.22). Force–armour VIF 8.4 → **2.1**. All 18 coefficients stay non-zero: ridge only *shrinks*.

Coefficient shrinkage path (schematic)

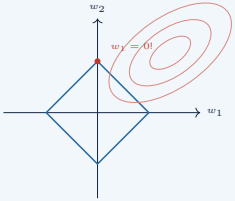


All coefficients decay smoothly toward 0 as λ grows; none hits exactly 0.

Lasso (L1) & Elastic Net — Sparsity and Feature Selection



Ridge: sphere, no corners



Lasso: diamond, corners on axes

LASSO AND ELASTIC NET

$$L_{\text{Lasso}} = \|y - Xw\|^2 + \lambda\|w\|_1$$

$$L_{\text{EN}} = \|y - Xw\|^2 + \lambda_1\|w\|_1 + \lambda_2\|w\|^2$$

- No closed form ($|w_j|$ not differentiable at 0): coordinate descent, ADMM, LARS
- L1 creates *corners* in the constraint region: the solution prefers to land ON an axis ($w_j = 0$) $\Rightarrow$ automatic feature selection

Case A: Lasso ($\lambda = 0.3$, 5-fold CV)

Test $R^2 = 0.728$, RMSE 4.02 h. **6 of 18 features zeroed**: comms quality, temperature, days-since-resupply, morale index, fog-of-war, enemy experience. 12 retained $\rightarrow$ simpler, deployable model.

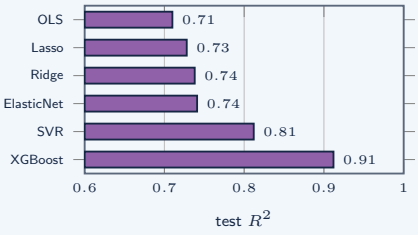
Elastic Net ($\lambda_1 = 0.2$, $\lambda_2 = 0.5$)

Best of both: sparse *and* stable with correlated features. $R^2 = 0.741$, **8 features zeroed** (10 used).

Ridge vs Lasso vs Elastic Net — When to Use Which, and the Full Case A Table

Method	λ (tuned)	Test R^2	RMSE (h)	Features	Train (s)
OLS (no reg.)	—	0.710	4.22	18/18	0.06
Ridge	0.8	0.738	3.91	18/18	0.07
Lasso	0.3	0.728	4.02	12/18	0.14
Elastic Net	0.2, 0.5	0.741	3.88	10/18	0.18
SVR (RBF)	C=50, ε=0.5	0.812	3.24	18/18	4.8
XGBRegressor	M=400, η=0.05	**0.912**	**2.06**	18/18	22.4

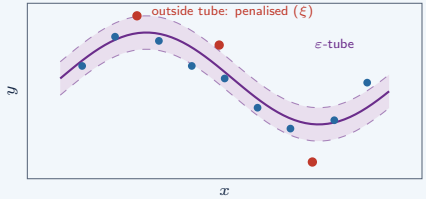
XGBRegressor dominates ($R^2 = 0.912$, RMSE 2.06 h). Elastic Net is best among linear methods: an interpretable 10-feature model with $R^2 = 0.741$.



Which penalty?

Ridge: default for collinear features, keep all. **Lasso:** when many features are irrelevant. **Elastic Net:** unknown structure — the recommended default; two hyper-parameters.

Support Vector Regression — Fitting Within a Tube: No Penalty for Small Errors



ε-INSENSITIVE LOSS AND PRIMAL

$$L_\varepsilon(y, \hat{y}) = \max\left(0, |y - \hat{y}| - \varepsilon\right)$$

$$\min \tfrac{1}{2}\|w\|^2 + C\sum_i (\xi_i + \xi_i^*)$$

- Constraints: $y_i - w^\top x_i - b \le \varepsilon + \xi_i$ (upper), $w^\top x_i + b - y_i \le \varepsilon + \xi_i^*$ (lower), $\xi, \xi^* \ge 0$
- C: tolerance for errors OUTSIDE the tube; large C = tight fit (may overfit)
- ε: half-width of the tube; larger ε = more points inside = sparser model
- Prediction $\hat{y} = \sum_i (\alpha_i - \alpha_i^*) K(x_i, x) + b$ (RBF kernel for Case A)

Case A grid search: $C = 50$, $\varepsilon = 0.5$, $\gamma = 0.05$: test $R^2 = 0.812$, RMSE 3.24 h; **1,842 support vectors (42.4%** of training data).

Benefit 1: robust to outliers

Training data has extreme battles (2-hour ops to 48-hour sieges). With $\varepsilon = 0.5$, prediction errors under 30 min are IGNORED — the model focuses on getting the large errors right.

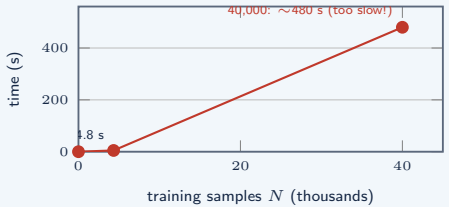
Benefit 2: non-parametric kernel mapping

RBF captures non-linear terrain–duration relationships linear methods miss: mountain terrain + rain is a *multiplicative* effect, not additive.

Benefit 3: sparse model via support vectors

Inference $\propto$ #SVs $\times$ kernel cost, independent of N once the SVs are found.

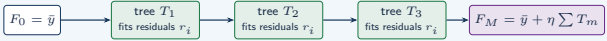
Training time vs sample size



Straight-line extrapolation from the reported 4,340 → 40,000 sample figures; kernel SVR is $O(N^2)$–$O(N^3)$, so real growth can be worse.

Trade-off: kernel SVR training is $O(N^2)$ to $O(N^3)$: 4,340 samples → 4.8 s is fine; 40,000 samples would take ∼480 s. Use **XGBoost** for large datasets.

Gradient Boosting for Regression — XGBRegressor: Sequential Residual Learning



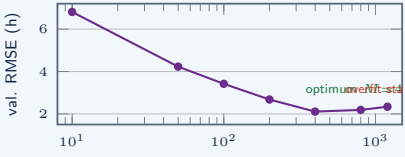
$r_{im} = y_i - F_{m-1}(x_i)$: what the model got wrong; trees predict residuals, NOT the target

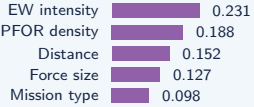
UPDATE AND XGBOOST OBJECTIVE

$$F_m(x) = F_{m-1}(x) + \eta\, T_m(x) \;\; (\eta = 0.05,\ \text{depth } 5)$$

$$\mathcal{L} = \sum_i (y_i - \hat{y}_i)^2 + \sum_m \Omega(T_m), \quad \Omega(T) = \gamma|\text{leaves}| + \tfrac{1}{2}\lambda\|w\|^2$$

Second-order Taylor expansion is used for speed. Hyper-parameters (5-fold CV): $M = 400$, $\eta = 0.05$, depth 5, subsample 0.8, colsample 0.8, $\alpha = 0.1$, $\lambda = 1.0$.





Final test: $R^2 = 0.912$, RMSE 2.06 h, MAE 1.54 h; importances consistent with the OLS coefficients.

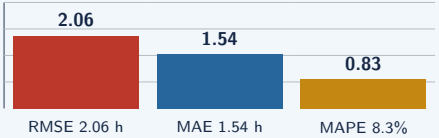
Regression Metrics — What They Measure and When They Mislead

THE FOUR CORE METRICS

$$\text{MSE} = \frac{1}{N}\sum(y_i - \hat{y}_i)^2, \quad \text{RMSE} = \sqrt{\text{MSE}}$$

$$\text{MAE} = \frac{1}{N}\sum|y_i - \hat{y}_i|, \quad \text{MAPE} = \frac{100}{N}\sum\frac{|y_i - \hat{y}_i|}{|y_i|}$$

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}, \quad \text{Adj.}R^2 = 1 - (1 - R^2)\frac{N-1}{N-d-1}$$

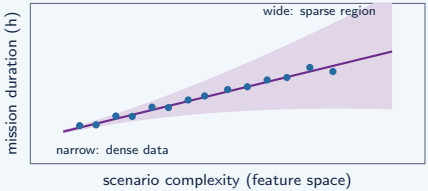


XGBoost, Case A. RMSE $>$ MAE confirms outlier battles exist. $R^2 = 0.912$: the model explains 91.2% of variance in mission duration. (MAPE bar scaled $\times 0.1$.)

Metric traps in defence regression

- **Trap 1:** high train R^2, low test R^2 (0.98 vs 0.71) = overfitting; report both (OLS: 0.738 vs 0.710)
- **Trap 2:** RMSE hides *directional bias*: missions taking LONGER than predicted is critical for logistics; report signed mean bias $\sum(y_i - \hat{y}_i)/N$
- **Trap 3:** R^2 can be *negative* if the model is worse than predicting the mean ($SS_{\text{res}} > SS_{\text{tot}}$)
- **Trap 4:** time-series exercise data inflate R^2 via autocorrelation $\Rightarrow$ use an out-of-time test
- **Trap 5:** MAPE is undefined when $y_i = 0$; use sMAPE or RMSE

Prediction Uncertainty — Knowing When Your Model Doesn't Know



OLS PREDICTION INTERVAL

$$\hat{y}_{\text{new}} \pm t_{N-d-1,\alpha/2}\, \hat{s}\sqrt{1 + x^\top (X^\top X)^{-1} x}$$

Why it matters

"Mission takes 18.4 h" is operationally incomplete. The commander needs "± what?", the 90% upper bound, "should I plan for 22 h?".

- **Aleatoric** (irreducible): random enemy behaviour, weather; not reduced by more data
- **Epistemic** (model ignorance): sparse training regions; reducible with more data

Practical intervals

RF/XGBoost: $\hat{y} \pm 1.645\,\sigma_{\text{pred}}$ from variance over $B = 500$ trees: average width ± 3.8 h. **Bayesian ridge** ($w \sim \mathcal{N}(0, \frac{\sigma^2}{\lambda} I)$, posterior $\mathcal{N}(w_{\text{Ridge}}, \sigma^2 (X^\top X + \lambda I)^{-1})$): avg 95% CI ± 6.1 h; 90% coverage **89.4%** (well calibrated); in sparse regions CI widens to $\pm 12+$ h.

Defence value: "18.4 h but ± 12 h for this mountain-jungle mission" beats blindly reporting 18.4 h with false precision.

POISSON REGRESSION (A GENERALISED LINEAR MODEL)

$$Y \sim \text{Poisson}(\mu), \qquad \ln \mu = \beta_0 + \beta_1 x_1 + \cdots$$

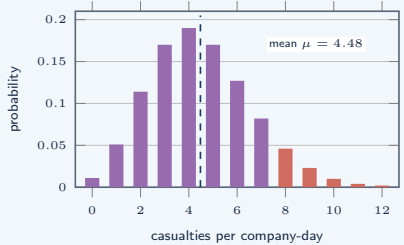
- log link keeps $\mu > 0$; effects are **multiplicative**: rate ratio e^{β_j}
- **offset** $\ln(\text{exposure})$ for unequal unit-days or strengths
- OLS would predict negative counts and assume constant variance

Worked example: casualties per company-day

Fitted on simulated engagements: $\ln \mu = 0.5 + 0.25\,x$, $x =$ enemy artillery battalions in range.
Rate ratio $e^{0.25} = \mathbf{1.28}$: +28% per battalion.
$x = 0$: $\mu = 1.65$; $\quad x = 2$: $\mu = 2.72$; $\quad x = 4$: $\mu = \mathbf{4.48}$

Over-dispersion: real casualty data have catastrophic tails (variance > mean). Check deviance/df; use the **negative binomial** model for honest, wider intervals.

Plan against the tail, not the mean ($x = 4$)



$P(\geq 8) = \mathbf{8.5}\%$; 95th percentile $= 8$. Medical and evacuation capacity should be sized for ≈ 8, not 4.5.

Casualty estimates are **planning inputs**, not forecasts of fact: report ranges with stated assumptions; the uncertainty must reach the commander.

LANCHESTER SQUARE LAW (AIMED FIRE)

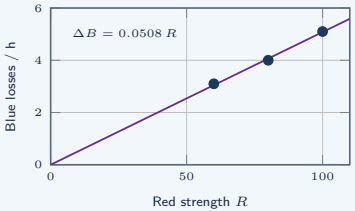
$$\frac{dB}{dt} = -r\,R, \qquad \frac{dR}{dt} = -b\,B$$
$$\Rightarrow\; b(B_0^2 - B^2) = r(R_0^2 - R^2)$$

Worked example: Red's kill rate from a simulation

Red strength R: 100, 80, 60 tanks; Blue losses ΔB: 5.1, 4.0, 3.1 per hour.

Regression through the origin: $\hat{r} = \dfrac{\sum R\,\Delta B}{\sum R^2} = \dfrac{1{,}016}{20{,}000} = \mathbf{0.0508}$ per Red tank per hour. Fitted 5.08, 4.06, 3.05 — consistent with the square law.

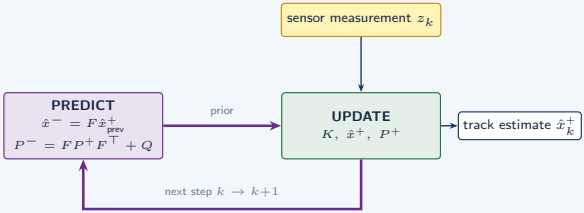
- **Which law?** Fit $\ln \Delta B = \ln c + p \ln R + q \ln B$: square law $p = 1, q = 0$; linear law $p = 1, q = 1$. Historical fits fall in between.
- Modern models regress effectiveness on terrain, weather, morale, EW: R^2 0.61 → 0.87 (L1 example).



Force-ratio planning: Blue prevails if $bB_0^2 > rR_0^2$, i.e. $\dfrac{B_0}{R_0} > \sqrt{r/b}$. If Red's kill rate is **4**× Blue's, Blue needs **2**× Red's numbers, not 4×.

$\sqrt{r/b}$ combines two uncertain estimates non-linearly: bootstrap the engagements and report an interval before using it in planning.

Kalman Filter — Predict, Measure, Fuse: Recursively & Optimally



STATE-SPACE MODEL

$$x_k = Fx_{k-1} + Bu_k + w_k, \quad w_k \sim \mathcal{N}(0, Q)$$

$$z_k = Hx_k + v_k, \quad v_k \sim \mathcal{N}(0, R)$$

UPDATE STEP (INNOVATION $\nu_k = z_k - H\hat{x}_k^-$)

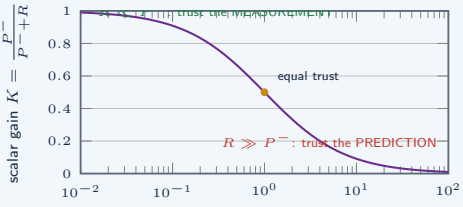
$$K_k = P_k^- H^\top \left(HP_k^- H^\top + R\right)^{-1}$$

$$\hat{x}_k^+ = \hat{x}_k^- + K_k\nu_k, \quad P_k^+ = (I - K_kH)P_k^-$$

- x_k: state (position, velocity, heading); F: transition; Bu_k: known manoeuvres; w_k: model error
- z_k: sensor measurement (radar return); H: maps state to observation; v_k: sensor noise

3-sensor Case B setup

State $x = [X, Y, \dot{X}, \dot{Y}]^\top$ for each of 5 targets. **Radar** $R = \text{diag}(50^2, 0.5^2, 10^2)$ (range ± 50 m, bearing $\pm 0.5^\circ$, speed ± 10 km/h). **IR** $R = \text{diag}(0.8^2, 1.2^2)$ (angle-only). **ELINT** $R = \text{diag}(0.1^2, 0.2^2, 5^2)$ (freq, PRF, bearing $\pm 5^\circ$).



- $K_k \to 0$ when $R \gg P^- H^\top H$: sensor noise exceeds prediction uncertainty $\Rightarrow$ trust the **prediction**
- $K_k \to H^{-1}$ when $R \ll P^- H^\top H$: trust the **measurement**, override the prediction

Illustrative 1-D worked update (not exercise data)

Prior $\hat{x}^- = 100$ km, $P^- = 4$; radar reads $z = 104$ km with $R = 1$.

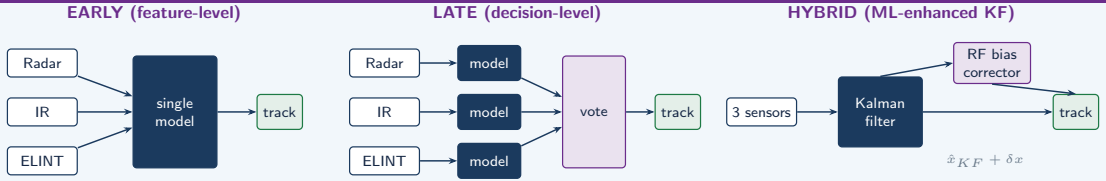
$K = \dfrac{4}{4+1} = 0.8 \quad \hat{x}^+ = 100 + 0.8(104 - 100) = \mathbf{103.2}$ km $\quad P^+ = (1 - 0.8)(4) = \mathbf{0.8}$.

A precise radar pulls the estimate 80% of the way toward the measurement and cuts the variance five-fold.



Fused estimate is narrower than either input: fusion always reduces uncertainty.

Multi-Sensor Fusion — Architecture Choices: Early, Late & Hybrid



Early fusion: 91.4%

✓ learns cross-sensor correlations; one model. × fails with missing data; sensitive to different sampling rates; high-dim input needs more data.

Late fusion: 89.7%

✓ robust to sensor failure (omit); different model per sensor; easy to add/remove sensors. × loses cross-sensor correlations; each sub-model needs full labels.

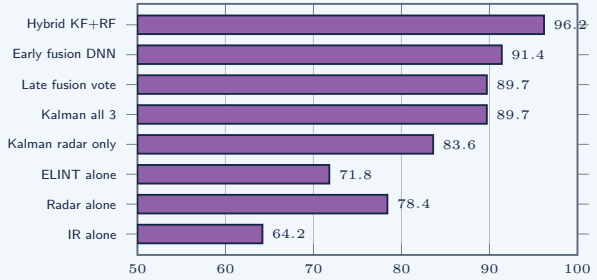
Hybrid: 96.2% — best

KF handles timing & physics; ML corrects systematic bias in KF residuals: $\delta x = g(\nu_k, \text{feat}_k; \theta)$, $\hat{x}_{\text{final}} = \hat{x}_{KF} + \delta x$. Physics-grounded, interpretable, handles rare events.

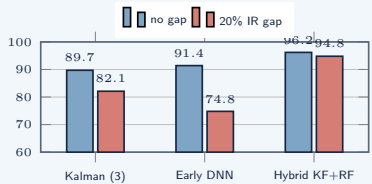
Recommended for IADS fusion: hybrid (ML-enhanced Kalman). A standard KF assumes linear dynamics and Gaussian noise — both violated in combat (evasive manoeuvres, jamming spikes); ML corrects both. Long pedigree in missile guidance.

Case Study B — 3-Sensor Fusion Results (5 Targets, 10 Runs)

Target-assignment accuracy by method (%)



Robustness: IR missing 20% of the time (cloud)



Method	Latency
Radar alone	0.3 ms
Kalman (all 3)	1.2 ms
Early DNN	3.8 ms
Late vote	4.1 ms
Hybrid KF+RF	**2.3 ms**

The hybrid loses only 1.4 points under the IR gap; the early-fusion DNN loses 16.6.

How the Hybrid Model Works — ML Corrects the Kalman Residuals

1 Kalman update
run the standard 3-sensor KF:
$\hat{x}_{KF}, P_{KF}, \nu_k$ per target

→

2 Residual features
f_k: innovation ν_k, $\|\nu_k\|^2$, S_k, Δt_k, sensor availability

→

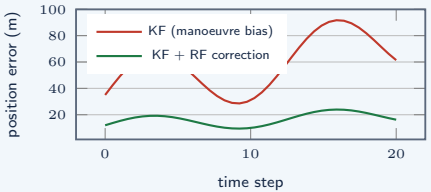
3 RF bias corrector
$\delta x_k = \mathrm{RF}(f_k)$, trained on true-minus-KF error pairs

→

4 Apply
$\hat{x}_{\text{final}} = \hat{x}_{KF} + \delta x_k$

Why residuals as features?

The innovation $\nu_k = z_k - H\hat{x}_k^-$ encodes how much reality differs from the KF prediction, sensor-specific bias patterns, and manoeuvre signatures the KF missed. $S_k = HP_k^-H^\top + R$ is the innovation covariance; Δt_k = time since each sensor last updated. The RF learns the systematic *non-Gaussian* patterns in ν.

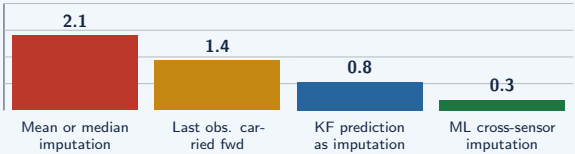


Schematic: RF removes the systematic manoeuvre-correlated component of the KF error.

Why sensors go missing

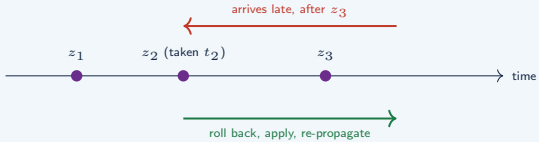
Radar: terrain dead zones, jamming (30–80% loss), ducting. **IR:** cloud, precipitation, clutter, vibration (15–40% degradation). **ELINT:** enemy EMCON, LPI emissions. In Case B, IR is blocked 20% of the time.

Accuracy impact of each imputation strategy



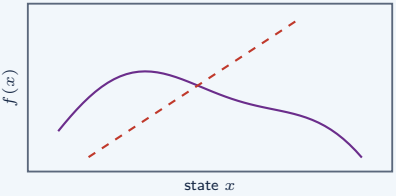
Accuracy loss in points (lower is better). Best: ML cross-sensor imputation (RF predicts IR from radar + ELINT).

Out-of-sequence measurements (OOSM)



Naive KF processes in arrival order: error. The radar update arrives 0.3 s after it was taken; retro-correct by rolling back to the measurement time, applying it, then re-propagating.

Data pitfall: naive imputation can CORRUPT fusion by introducing *false precision* — a filled-in IR bearing looks as certain as a real one (the KF assigns it the same R). Track imputation flags and **inflate** R for imputed measurements.



Extended Kalman Filter (EKF)

Standard KF: linear dynamics only. EKF linearises $f(x)$ around the current estimate: $F_k = \partial f/\partial x\big|_{x=\hat{x}_{k-1}}$ (the Jacobian) and replaces F at every step. Applicable to aircraft evasive manoeuvres; truncation error from linearisation.

Unscented Kalman Filter (UKF)

Propagates a small set of carefully chosen **sigma points** through the non-linear function exactly — no Jacobian. Better than EKF for strongly non-linear dynamics: hypersonic target tracking, UAV swarms.

Combat violates both KF assumptions: dynamics are non-linear (EKF/UKF) and noise is non-Gaussian (ML bias correction).

Case A: mission-duration regression ($N = 6{,}200$, 18 features)

Model	R^2	RMSE	MAE	Train (s)	Feat.
OLS	0.710	4.22	3.12	0.06	18/18
Ridge	0.738	3.91	2.88	0.07	18/18
Lasso	0.728	4.02	2.94	0.14	12/18
Elastic Net	0.741	3.88	2.81	0.18	10/18
SVR (RBF)	0.812	3.24	2.37	4.8	18/18
XGBRegressor	**0.912**	**2.06**	**1.54**	22.4	18/18

Case B: 3-sensor fusion (radar + IR + ELINT, 5 targets, 10 runs)

Method	Acc.	Latency	20% IR gap	Notes
Radar alone	78.4%	0.3 ms	—	baseline
Kalman (3)	89.7%	1.2 ms	82.1%	standard
Early DNN	91.4%	3.8 ms	74.8%	fragile
Hybrid KF+RF	**96.2%**	2.3 ms	**94.8%**	best

Case A winner: XGBoost ($R^2 = 0.912$, RMSE 2.06 h). **Case B winner:** Hybrid KF+RF (96.2% accuracy, robust to sensor dropout). In both cases the best system combines a strong learner with domain structure: tree ensembles for tabular drivers, a Kalman backbone for physics.

From Galton's peas to Apollo's navigation computer — and on to the wargame

BIOLOGY / STATISTICS
Galton (1886): children of very tall parents are tall but less extreme — "regression towards mediocrity" gave the method its name.
Defence: exercise scores of the best and worst units drift to the mean; don't credit (or blame) interventions without a control group.

AERONAUTICS
The Kalman filter was developed for Apollo navigation at NASA Ames (Schmidt, early 1960s) to fuse noisy measurements on board.
Defence: now in missile seekers, INS/GPS navigation and every radar track manager (Case B).

ENGINEERING
Fatigue S–N curves regress log-cycles-to-failure on stress amplitude to set component life limits.
Defence: airframe life and spares forecasting — and the danger of extrapolating beyond tested loads.

PSYCHOLOGY
Yerkes–Dodson (1908): performance rises then falls with arousal — an inverted U that a straight line cannot fit.
Defence: operator workload and stress models need non-linear terms (SVR, boosting), not OLS.

OPERATIONS RESEARCH
Engel (1954) fitted Lanchester's square law to the Iwo Jima battle record — regression as validation of a combat model.
Wargaming: calibrating attrition coefficients from exercise data (cf. ML-enhanced Lanchester, Lecture 1).

BIOLOGY → CBRN
Dose–response curves (logistic / probit regression) relate exposure to effect in pharmacology and toxicology.
Wargaming: CBRN casualty estimation from agent concentration and exposure time.

Common thread: regression estimates a *relationship*, not a cause; its predictions are trustworthy only inside the range of the data it was fitted on, and always need an uncertainty band.

Lecture 3 Summary — Regression Algorithms & Sensor Fusion

OLS, RIDGE (L2) & LASSO (L1)
Ridge shrinks; lasso zeros coefficients; elastic net combines both; λ by cross-validation.
Use: ridge for collinear features, lasso for sparse

SVR (ε-tube)
Ignores small errors; robust; kernel extends to non-linear; slow for $N > 10{,}000$.
Use: medium datasets, non-linear fits

XGBOOST REGRESSOR
$R^2 = 0.912$, RMSE 2.06 h; sequential residual learning; fast inference.
Use: default choice for tabular regression

KALMAN FILTER
Predict–update recursive cycle; optimal for linear Gaussian; EKF/UKF for non-linear.
Use: physics-grounded tracking baseline

HYBRID KF + ML
96.2% tracking accuracy; KF physics + ML bias correction; robust to 20% sensor dropout.
Use: best sensor-fusion architecture

UNCERTAINTY & METRICS
Report intervals, not points; R^2 with RMSE and signed bias; watch the five metric traps.
Use: always, before fielding

Lecture 4: finding structure in unlabelled combat data

- Why cluster in defence: discover unknown enemy tactical patterns without labels
- k-Means: derivation, elbow + silhouette selection, Lloyd's convergence
- DBSCAN: core/border/noise points; noise handling for cluttered sensor data
- Gaussian Mixture Models: EM algorithm for soft assignments; overlapping manoeuvre patterns
- Hierarchical clustering: dendrograms for analyst visualisation
- Dimensionality reduction: PCA (eigenvectors of the covariance), t-SNE / UMAP

Assignment 3: regression derivations + Kalman-gain computation; solutions provided for self-check.

Preview: OPFOR tactical archetype discovery

$N = 8{,}000$ force-movement trajectories (JANUS, 5 years); 12 kinematic + contextual features. k-Means ($k = 5$): silhouette **0.712**; 5 archetypes: patrol, assault, transit, static defence, **infiltration**. DBSCAN ($\varepsilon = 0.45$, MinPts 8): 4 clusters + **142 noise points (1.8%)** — deception manoeuvres(!). GMM ($K = 5$, AIC/BIC): log-likelihood $-24{,}816$. PCA: first 4 PCs explain **84.3%** of variance; t-SNE 2-D maps for situational-awareness displays.